PORTO UNIVERSITY

Faculty of Engineering of the University of Porto

Sub - Group discovery and Meaningful Distribution Rules in Time Series that lacks information on forecast wine production

Thesis preparation Report

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Contents

Abstract

This is a first report about the progress of my thesis. The main objective is to give a status essential about the literature reviewed at the current stage. The literature was manly about the business related with the Wine Production Volumes and how is this related with meteorological variables.
The main goal of this preliminary work was also to be able to define more concretely the research question to be addressed and also the macro steps needed to apply the methodologies founded.
As a work within the scope of the Thesis preparation unit, this is a work in progress document as it's expected that some of the content within the following chapters can be updated based on following research.

Keywords: *wine production, meteorological data, set group discovery, Distribution Rules*

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1 Introduction

Wine production is characterised by large inter-annual fluctuations. Climate regulates nearly every step of Wine Production Volume (WPV)

Detecting and characterising changes of the dynamics involved in WPV over time is the natural first step towards identifying the key determinants factors of this process.

The link between wine production and in-season climate data has been studied through many research articles, but it known that past season's climate have a strong impact too. Climate, has a general and directly affect, but as production technology changes, there is a non-constant impact into WPV, which translates into changing relationships between WPV and climate.

Therefore the detection of structural changes in grapewine yield and WPV time series should take this into account, especially in models where climate plays an important and direct or indirect role [Quiroga and Iglesias(2009), Cunha and Richter(2016)]

These effects combined with long-term (technological and climate) effects on WPV makes very difficult the understanding of WPV dynamics and could explain the lack of well stabilized and generalized models and methodologies to support the wine industry. [Cunha and Richter(2020), Cunha and Richter(2016)]

Therefore, is of great interest and a challenging problem to research about it. As we will see along this work, wine production is a multi-year process that is influenced by a range of factors over time. In addition to climate conditions during the grapevine flower formation stage, there are other factors that can influence the quality and quantity of grapes, and therefore the wine production, include:

1. **Viticulture practices:** This includes practices such as pruning, soil management, which can impact the health and productivity of the grapevines.

1. **Weather patterns:** Weather conditions during the growing season, including temperature, rainfall, and other's, can affect the growth and maturation of the grapes.

2. **Grape variety:** Different grape varieties have different growth patterns, maturation periods, and wine-making characteristics, which can impact the wine production.

3. **Vineyard location:** The geography, topography, and soil type and soil water of a vineyard can also have a significant impact on the wine production,

4. **Harvest timing:** The timing of the harvest is also an important factor in wine production, as it can impact the quality and flavor of the grapes.

These are just a few of the factors that can influence WPV over the course of several years.

The process of wine production is complex and requires careful planning and management to ensure that the grapes are of the highest quality and the wine is of the best possible flavor and character, and also the maximization of the volume produced in each year is achieved.

Climate conditions during the grapevine flower formation stage can greatly impact the quality and volume of grapes. We can think that, for instance, the temperature is important in specific stages of the grapes development but also important for maintain or create soil water conditions.

Increased temperatures are likely to be associated with increased water loss in the vineyard.

Rainfall is also linked with the previous variables, and an sweet point is needed, because the lack or the excess of rain in certain periods of the grapes development cycle, can have different (positive or negative) impacts in the wine production cycle.
All this variables are in some way linked with each other and in a complex manner. Be able to understand this relations, or derive some explanations or rules that help's a better understanding is a major contribution to the wine producers and related organizations.

1.1 Challenge of vineyard yield variability for yield prediction

There are many factors that can influence grapevine growth and fruit production.
They can vary across different vineyards, even within the same region. Some of them include climate, soil type, grape variety, vine age, pruning techniques, and others.

Grapevines can be highly responsive to environmental conditions, and as for, this relation can lead to a high degree of yield variability with respect to those factors. To address the variability challenge, wine producers often use a combination of historical data about past productions for prediction proposes.
Often, meteorological data isn't included within the model's for several reasons, from difficulties in gather the data, to the lack of technical expertise to extract the knowledge.
Wine production is one of the businesses where the impact of climate change is particularly noticeable [Cunha and Richter(2016)] and reasoning about last climate changes over the world, majors changes and impacts are expected to occur into the grapevine and consequently higher variability on the WPV.
Related with grape variety, recent adoption of common high-yielding grape varieties and uniform agronomic practices (e.g. irrigation) have caused grape-yields to become more strongly influenced by weather patterns, especially under the foreseen climate scenarios.[Cunha and Richter(2020)]

Seasonal differences in weather conditions cause marked variation in grapevine yield. In [Zhu et al.(2020)] study, the relationships between various yield components and climatic conditions during critical periods are quantify and the correlations between each yield component and different weather factors is addressed. The idea is to try to explain the variations in different yield components based on weather conditions during critical periods and consequently maybe explain some of the vineyard yield variability.
As show in [Cunha and Richter(2020), Cunha and Richter(2016)], in Portugal wine regions, despite several improvements in vineyard technology, wine-yield remains highly dependent on the short and long-term climate, which causes important variations in wine production with several adverse effects.
Showed in [Cunha and Richter(2016)], wine production has always been volatile in terms of output variance.

From a technical perspective, regarding the time series about WPV in the Douro region from 1933 to 2013 evidence of an upward trend in WPV is showed. This time series contains changes in its typical cyclical behaviour with increasing variance, which may be caused by structural breaks, which, hypothetical, be caused by climate effects. Quantitative relationships between various yield components and climatic factors at field scales are still lacking. [Zhu et al.(2020)]

1.2 Why we need to predict wine production?

In a common sense approach, WPV prediction can be useful for a variety of reasons, including, purchasing reasons, assist wine producers in making decisions about grape harvesting, fermentation, aging, blending, bottling, processes optimization, wine appraisal for value estimation of a wine based on its quality, rarity, and age, etc. Thus, maintaining stable yields from year to year is essential for achieving a consistent fruit quality and supply, which are crucial within the context of increasing competition in the international market.[Zhu et al.(2020)]

In a more global sense, Governmental agencies, the wine industry and researchers have become more alert to the problems caused by climate variability and are looking for ways to manage them [Quiroga and Iglesias(2009), Cunha and Richter(2016)].
Be able to predict WPV is directly engaged with the ability to understand the possible climate factors and variables that influence the production volumes. Be able to find the hidden rules or patterns, it will not only increase the models performance but also help this Governmental agencies and wine industry to anticipate some of the impacts on WPV and acting strategical to minimize or bypass them.

[Cunha and Richter(2020), Cunha and Richter(2016)] point that, the drier regions of the Mediterranean basin, where the foreseen climate scenarios pointed for substantial drying, expect precipitation reductions of more than 25% and warming by 3-5 ºC by 2080. Long-term analysis of WPV would have a key importance, at the light of the this recent global and regional climate change scenarios, in order to plan future mitigation actions for viticulture and wine industry.

[Quiroga and Iglesias(2009)] also explain that, the temporal variability of WPV is a major point of interest for farmer and its associations, wine sellers, insurances, researchers, natural resources managers and policy makers.

Accordingly to [Cunha and Richter(2020), Cunha and Richter(2016)], there is a strong demand for study's and improvements regarding time series of regional wine production in order to improve the efficiency of vineyard and winery operations as well as to support commercial strategies and sectorial planning measures

Several articles state that, production levels , may be further impacted by climate change in the future despite noticeable advances in vineyard technologies,[Cunha and Richter(2016)], thus, more variability will be implicated and the need for reliable predictions would be required.
Nevertheless, the importance of grapevine yield on various viticultural and winemaking practices is identified, but little effort has been made to predict yield [Zhu et al.(2020)] and thus, this study aims to contribute to the discover of features to take into account for new models to arise in the future

1.2.1 Research question

This work has the objective to relate two distinct data mining approaches to find patterns and rules that could help to describe the complex relation between wine production and the meteorological variables such as climate measures.
How does meteorology affects WPV, is the main question to be investigated.

The idea is to apply Subgroup Discovery (SGD) techniques and Meaningful Distribution Rules (DR) approaches to explain how this meteorological variables influence one variable, the WPV for a certain year.

DR are related to quantitative association rules but can be seen as a more fundamental concept, useful for learning distributions. SGD is a descriptive induction technique that extracts interesting relations among different variables with respect to a special property of interest known as target variable.
Thus, our research question is:

Can tasks such as frequent pattern discovery, sub group discovery and features extraction be derived from the time series?

From an academic perspective, we found a robust methodology to find certain distribution rules. We will use a know framework from [Jorge et al.(2006)].

For the SGD part, research will be done during the thesis elaboration, because some approaches were founded, but it seems to be missing a clear approach to deal with the fact that, the problem that we are dealing it is a relation between features that are indeed time series (daily meteorological data), and that, in some way, influence a single point prediction (the WPV for a certain year) in a particular moment in time.

1.2.2 Problem definition and objectives

The objective is to understand how the meteorological variables influence the WPV of a certain year.
The WP time series are not stationary and include transitory components, consisting of a variety of frequency regimes that may be localized in space or have short-lived transient components and features at different scales [Cunha and Richter(2016)]. Stationary means not only, that a series has a constant mean and constant and finite variance, but also that the autocovariances are not functions of time. [Cunha and Richter(2020)]

2 Literature Review

In the next sections, a literature is presented, in a way that divides the subjects into two distinct classes. The first one is about the theoretical main specifications about the business and the second one about the technical approach's that will be used along this thesis.

Because of the high importance of a very good understanding about the development cycle of the grapes, some attention is given to the grapewine yeld formation and the meteorological impacts in the wine yeld. The wine regions used are very distinct, nevertheless, being geographically close to each other in terms of distance. Some explanation about each one is also given.

Related with the technical aspect of the work, some overview about the fundamentals of distribution rules is describe and for subgroup discovery, an overview of the state of the art is presented with a clear intention of showing an initial path to research about it.
Our goal is to research deeper about the methodologies that exists and derive one possible approach for our research question. That will be done along the second part of the thesis work.

2.1 Main concepts

Some of the concepts used in this work are related with wine technical fields and wine slang used in the business. Here we present some of them for a better understanding of he content.

- **Wine yield** refers to the amount of grapes that are harvested from a vineyard and used to produce wine. It is usually measured in terms of tons of grapes per acre or hectare of land. Wine yield can vary depending on several factors, including weather conditions, soil quality, grape variety, and vineyard management practices. The amount of wine yield can have a significant impact on the overall quantity and quality of wine produced. In general, higher yields can result in lower quality wine, while lower yields can result in higher quality wine with more concentrated flavors and aromas. Winemakers and vineyard managers often strive to balance the desire for higher yields with the need to produce high-quality wine.

- **Vineyard yield** refers to the total amount of grapes that are produced in a vineyard in a given growing season. It is typically measured in tons or pounds per acre, hectare, or other unit of area. Vineyard yield is influenced by several factors, including grape variety, climate, soil type and quality, vineyard management practices, and pest and disease pressures

- **The spatio-temporal variability of wine yield** refers to the variation of grape yields across space and time in a vineyard or a wine-producing region. The yield of grapes can vary significantly within a single vineyard or region due to a number of factors, such as differences in soil type, drainage, sunlight exposure, and microclimates. These factors can affect the timing of flowering, the development of grape clusters, and the overall health of the grapevines, all of which can impact the final yield

- **Temporal variability in wine yield** refers to changes in the yield of grapes over time due to factors such as climate variability, weather events, pest and disease pressures, and changes in vineyard management practices

- **Climate** refers to the long-term pattern of weather conditions in a particular region or location. It includes measurements of temperature, humidity, precipitation, wind, and other atmospheric factors that occur over many years. Climate is typically defined in terms of the average conditions of a region, as well as the seasonal and inter-annual variations in those conditions

- **Meteorological** refers to the study and analysis of weather and weather patterns. Meteorological data is collected through a network of weather stations and sensors that measure these variables at different times and locations

- **"Anlagen"** is a German word that is often used in the context of wine production and refers to the specific growing conditions that exist in a particular vineyard or wine-growing region. These conditions can include things such as the climate, soil type, altitude, and exposure to sunlight, as well as other factors that can influence the quality and character of the grapes that are grown in that area. In terms of wine yield, the "Anlagen", or growing conditions, can play a significant role in determining the quantity of grapes that a vineyard can produce. For example, a vineyard located in a warm and sunny climate with fertile soil and adequate water resources is likely to produce a larger crop than a vineyard located in a cooler and more challenging environment. Additionally, the way in which the vineyard is managed and cared for, such as the pruning techniques used and the type of trellis system employed, can also impact the yield of the vineyard.

- **Predicting**, in the context of time series analysis, refers to the process of forecasting future values of a variable based on historical observations of that variable.

- **"Véraison"** is a term used in the wine industry to describe a critical stage in the development of grape berries. "Véraison" is the point at which the grapes start to change color and mature, marking the transition from the growth stage to the ripening stage. During "Véraison", the berries will change color from green to their final color, which could be red, black, or blue,depending on the grape variety. In terms of wine yield, is an important indicator of the potential yield of a vineyard. The timing and rate of "Véraison" can impact the final size and quality of the grapes and, in turn, the quality of the wine that is produced. For example, grapes that experience "Véraison" at an earlier time will generally have more time to ripen and may produce a larger crop, while grapes that experience "Véraison" later in the season may have less time to ripen and may produce a smaller crop. Additionally, the way in which the vineyard is managed during the "Véraison" period, such as the amount of water and nutrients provided and the pruning practices used, can also impact the yield and quality of the crop

- **Flowering** refers to the process by which the grape vine produces flowers, which will later develop into fruit. Flowering and "Véraison" are two important stages in the development of grape vines and the production of wine.The first refers to the process by which the grape vine produces flowers, which will later develop into fruit. This process usually occurs in the spring and is triggered by warm temperatures and increasing daylight hours. The number of flowers that are produced during flowering can have a significant impact on the potential yield of the vineyard. The second is the stage that occurs after flowering, when the grapes start to change color and mature. During "Véraison", the grapes will change color from green to their final color, which could be red, black, or blue, depending on the grape variety. "Véraison" marks the transition from the growth stage to the ripening stage and is an important indicator of the potential yield and quality.

- **Time series** is a sequence of data points that are collected at regular intervals over time. It is a statistical data analysis technique that focuses on analyzing and understanding the behavior of a variable or phenomenon over time. Time series data can be continuous, discrete, or categorical and can be measured on a wide range of scales, including hourly, daily, weekly, monthly, quarterly, yearly, or any other regular time intervals

- **Yield** per hectare is the product of vines per hectare, shoots per vine, inflorescence's per shoot, berry number per bunch and berry mass[Zhu et al.(2020)]

2.2 Regional Wine Production

Grapevines are one of the most important perennial crops, growing in over 30 different denominations of origin, which support a reputed wine industry strongly linked to specific wine regions.[Cunha and Richter(2020)].

Portugal is home to several wine regions that are known for producing high-quality and unique wines Some of some of the most notable wine regions in Portugal are: Douro (DR), Vinhos Verdes (VVR) , Alentejo, Dão, Bairrada and Colares (Lisbon). 1

Some characteristics of the more well known regions are:

- **Douro Region** - located in the northeast of Portugal and is considered one of the oldest and most prestigious wine regions in the world. he region is famous for producing Port wine, a fortified wine made from grapes grown in the Douro valley. Is also is one of the most arid wine regions of the world, with strong and consistent post-flowering water shortages

- **Vinhos Verdes Region** - located in the northwest of Portugal, stretching along the Atlantic coast from the Minho River to the Douro River. The region is known for producing young, light-bodied and refreshing wines.

- **Alentejo Region**: located in the south-central part of Portugal and is known for producing full-bodied and flavorful wines. The region's climate is warm and dry, with hot summers and cold winters.

- **Dão Region**: located in the center of Portugal, surrounded by mountains and forests. The region is known for producing elegant and complex wines

Regarding the two regions used in this work, DOuro region (DR) and Vinhos Verdes Region (VVR), in 2018, they represent at about 32% of Portugal' vineyard area and 37% of the wine production.
The overall differences about the difference in wines produced in this two regions is because their location and also the climate.
The average temperature for the year in VVR ranges between 12.5 and 15.8 °C, and in DR is 11.8 and 16.5 ºC. For both regions, the warmest month, on average, is August with an average temperature of 20 ºC (VVR) and 24.4 ºC (DR), and the coolest month on average is January, with an average temperature of 8.8 °C (VVR) and 8.0 ºC.
The average amount of annual precipitation varies from 1200 to 2000 mm in VVR and 400 to 800 mm in DR, most of which occur in autumn and winter. According to Köppen's classification, VVR belongs to the climate type Csb, while the climate in DR is Csa type. It is characterized by mild climate with dry and hot summers and has four months of the year when temperatures do not fall below 10 °C. The Csa is mild climate with dry and hot summers, with average temperatures in the hottest month exceeding 22 °C. [Cunha and Richter(2020)]

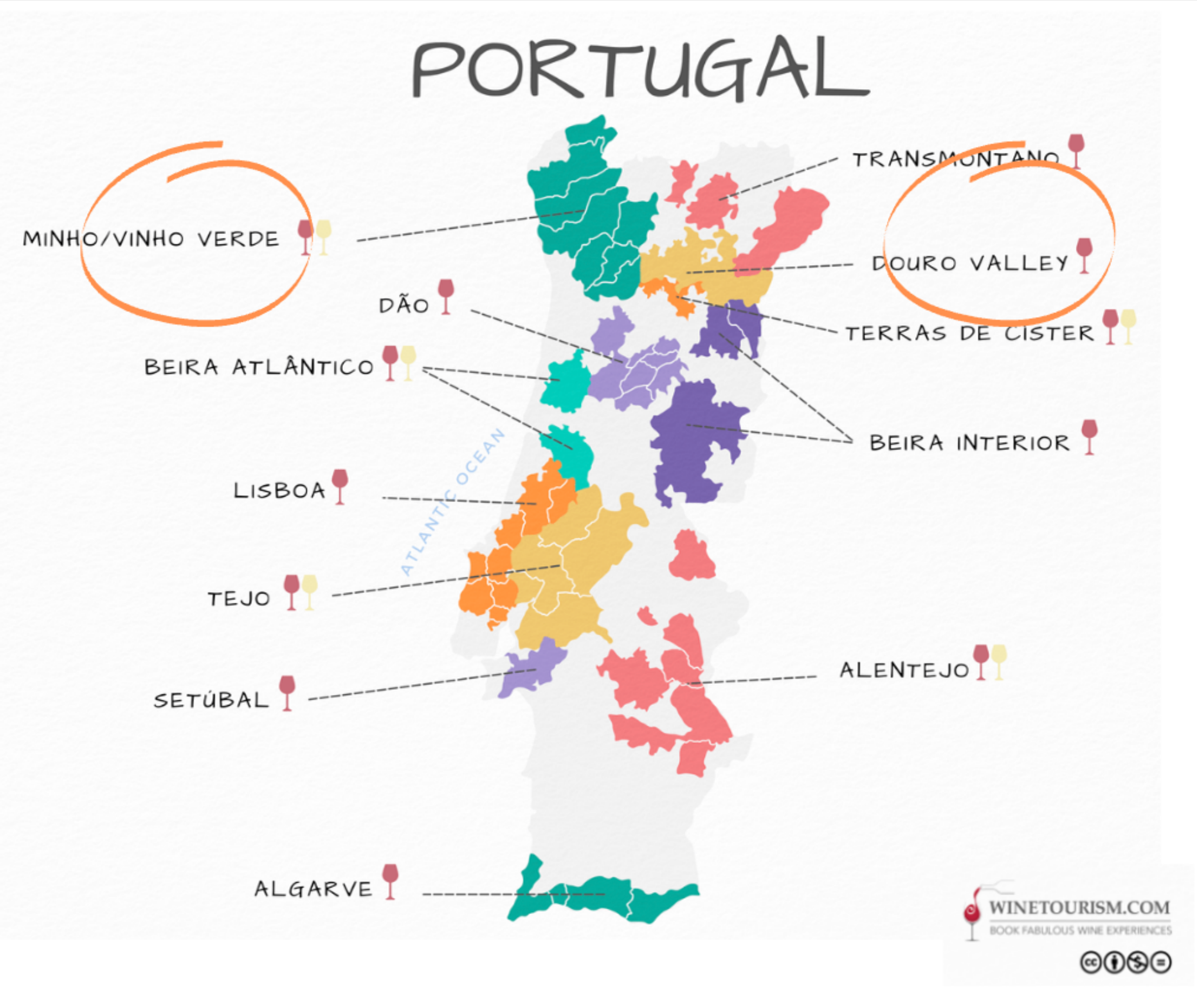

Figure 1: Portugal Wine Regions.

Figure adapted from [Tim(2022)]

2.2.1 Douro Region

The Douro wine region is located in the northeast of Portugal and is considered one of the oldest and most prestigious wine regions in the world. The vineyards are mainly built over terraces and slopes with soils mostly derived from shale. Also, mostly in the eastern part of this region, the vineyards are located in some of the most arid, non-irrigated, regions of Europe.
It is one of the most arid wine regions of the world, with strong and consistent post-flowering water shortages, hence, if climate change has an impact on dynamic behaviour of WPV.[Cunha and Richter(2020)]

The region covers an area of approximately 250,000 hectares, with the Douro River running through the heart of the region. The Douro wine region is famous for its production of Port wine which is a fortified wine made from grapes grown in the Douro valley. Is responsible for about one fourth (23%) of all wine produced in Portugal.
This region has a long history of wine making, back to Roman times. It is a demarcated region. It is a growing region in terms of tourism also due to the popularity of Port wine.

The DR has a unique climate and geography, with steep terraced vineyards that are planted on the hillsides

overlooking the Douro River. The soil in the region is predominantly schist, which is a type of metamorphic rock that is ideal for growing grapes.
It is also one of the most arid wine regions of the world with strong and consistent post-flowering water and thermal stress. Since most of the vineyards in the Douro region are non-irrigated and most of the rain occurs outside the vegetative growth cycle, soil water (SW) should be an important factor of the temporal variation of WPV in Douro. [Cunha and Richter(2016)]

In the DR, prevails non-irrigated vineyards, thus, the SW of the previous two and three years have a positive effect on WPV, suggesting that the precipitation during the rainy period (mainly autumn and winter) is not always enough to fill up the soil water reserve.[Cunha and Richter(2020)]
As in many other wine regions, the last decades were characterized by a large inter-annual fluctuation in wine [Cunha and Richter(2016)]
There is also a stable but not constant link between the production and the summer temperature (ST). [Cunha and Richter(2020)] refers that temperature is still important for explaining about 60% of the long term and short term behaviour of WPV.

Notice that, in projected climate changing scenarios for Douro region, soil water will be the limiting resource for wine production and thus, understating the relation within WV and SW within the region could be extremely important for defining strategy's.

2.2.2 Vinhos Verdes Region

The Vinhos Verdes Region (VVR) is located in the northwest of Portugal, stretching along the Atlantic coast from the Minho River to the Douro River. The region covers an area of approximately 34,000 hectares and is known for producing young, light-bodied and refreshing wines.

The wines produced in the region are meant to be consumed young and are not aged for long periods. The climate in the VVR is influenced by the Atlantic Ocean, with cool temperatures and high rainfall throughout the year. The region is the most Atlantic European wine region. Grapevines grown here have unique characteristics, namely, the form of guiding systems with wide vegetative expansion and growth high above the ground.[Cunha and Richter(2020)]
In this region, the rain occurs typically during the autumn and winter periods and are generally enough to refill the SW, avoiding carryover effects of water deficits that affects the WPV in subsequent years.[Cunha and Richter(2020)]

2.3 Grapevine Yield Formation

Grapevines commence forming their inflorescence's (floral induction) during the previous seasons and this process is highly influenced by the Soil Water (SW) level. The grape and crops cycle in wine production is the series of events that occur from the first stage of planting the grapevines to the final stage, the harvest and production of wine.

The general overall process, from a chronological perspective, can be described as having mainly 6 steps and can last several years from the initial stage:

1. **Preparation:** The soil is prepared, and a suitable location is selected based on factors such as climate, soil type, and topography.
2. **Planting:** Grapevines are planted, typically in late winter or early spring
3. **Training and pruning:** Over the next few years, the grapevines are trained and pruned to shape the canopy and encourage healthy growth.
4. **Bloom and fruit set:** In spring, the grapevines bloom, and the flowers are pollinated, leading to the formation of small berries.
5. **Grapevine development:** Over the summer months, the berries grow and mature, developing their flavor and aroma.
6. **Harvest:** It is the harvested of the grapes and happens in the fall

One of the important aspects in this work, is to understand which moments of the entire development cycle could be seen as important in terms of impacting the grapes growth. Some of the variability seen in the wine production annual values could be explained by some combination of specific mineralogical phenomenons at specific periods of the cycle. That means, for example, that, the fact that the entire cycle is very long, some past information on previous years could be impacting in same manner the particular year WPV.

Accordingly to [Cunha and Richter(2020)], to the perennial nature of grapevine wherein cluster structures present in one year begin their development in the spring/summer of the previous year. The spring includes a period of high intensity of growth and the reproductive phase between flowering and fruit-set which are physiological dependent of temperature, in this particular case, of the Spring Temperature. On the other hand, the Soil Water (SW) period includes the vegetative phase between "veraison" and harvest, characterized by strong potential hydric stress.

Grapevines may generally form up to four inflorescence's on a shoot and "anlagen" define the potential bunches for the following season.[Zhu et al.(2020)]
Grapevine initiate inflorescence primordia in the summer of the year prior to that in which they flower 2.

Accordingly to [Zhu et al.(2020)] whether or not "anlagen" become inflorescence or tendril primordia could be think of as an indicator of how much a productive harvest would be in the next year.
This enables us to anticipate potential yield early in the season, allowing adjustments of crop load to be made during winter pruning and after fruit-set. [Zhu et al.(2020)]
The behaviour of this cycle, will depend on temperature and radiation conditions around the bud at the time differentiation takes place. High temperatures during initiation will encourage inflorescence primordia development, while shaded or cool conditions will lead to tendril formation.[Zhu et al.(2020)]

Accordingly to [Zhu et al.(2020)] WPV is directly related with the number of bunches (inflorescence's) and berries in grapevine.
Flower development, which determines the number of bunches and berries , involves three main steps:

(1:) formation of "anlagen" or uncommitted primordia,

(2:) differentiation of "anlagen" when forming inflorescence or tendril primordia,

(3:) differentiation of flowers.

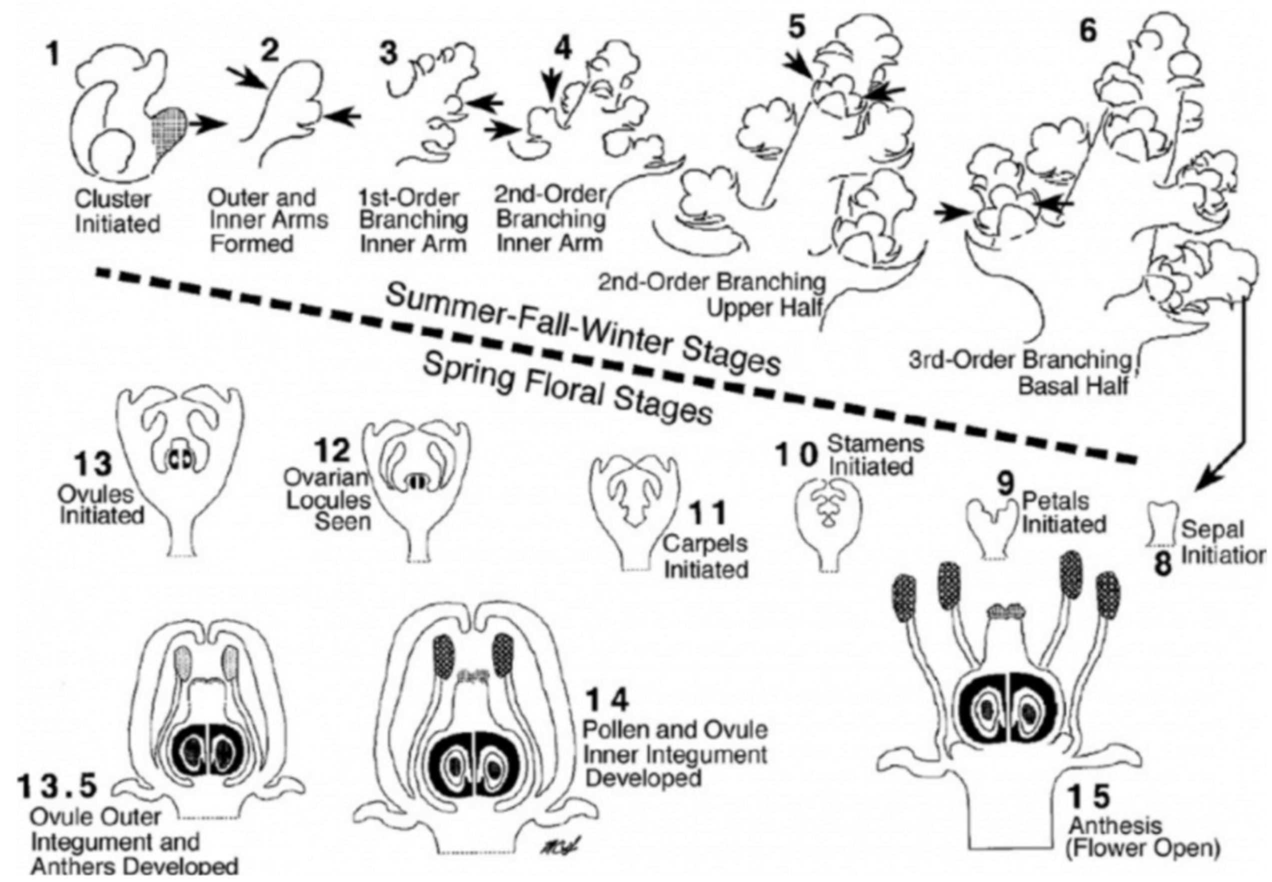

Figure 2: Graphical Representation of Grapevine formation cycle
Figure from [Tim(2022)]

2.4 Meteorological impacts on wine yield

Meteorological factors play a crucial role in determining the yield and quality of wine grapes. The success of wine production is heavily dependent on a variety of weather conditions that impact the growth and ripening of grapes.
In general knowledge, an intuition among wine producers and enthusiasts, temperature, rainfall, sunlight, and wind are some of the key meteorological factors that can affect wine grape yield.
For example, temperature has a significant impact on flowering, and ripening of grapes. Extreme heat or cold can damage the vines, while moderate temperatures are necessary for optimal growth and development.
Another example, although rainfall is essential for grapevine growth as for several agriculture productions, excessive rainfall can cause disease and reduce the wine yield. Also sunlight is required for photosynthesis, which is necessary for the production of sugars and flavors in the grapes.

Weather events during key development periods, the ones identified above that affect bunch number and berry number per bunch, have a direct an strong influence on wine yield. Conditions during the grapevine, flowering and inflorescence initiation are critical because not only, they affect the current season's berry number and mass, but also affect the following season's bunch number per vine, [Zhu et al.(2020)]. In the same work, the authors quantified the correlation between grapevine yield components and weather conditions during the key developmental stages by carrying out a long-term yield monitoring trial. They found that daily maximum temperature played a critical role in inflorescence initiation, while both daily maximum and minimum temperature played essential roles in berry number and berry mass.

Another confirmation already stated is that the perennial nature of the grapevine and its permanent structures are highly influenced by the past years' agroclimatic conditions. [Cunha and Richter(2016)]. Thus, current-year production is expected to be physiologically dependent on past weather conditions and water deficits. [Cunha and Richter(2020)]
Also, [Zhu et al.(2020)] show that seasonal or inter-seasonal factors like temperature and water, have pronounced effects on the fruitfulness and overall yield of grapevine.
The proportion of flowers that are retained as berries (Fruit set) is highly affected by cold weather after budburst as it may prevent the formation of individual flowers.
Both low and high temperature can affect fruit set in grapevines, and pre-flowering low temperatures can disrupt both the formation and function of ovules and pollen during flowering. [Zhu et al.(2020)]

As reference, the optimal temperature for growth, is assumed to be 22 °C for the period before budburst and 28 °C for the growth after budburst accordingly to [Zhu et al.(2020)].
Regarding the Portugal Wine Regions of interest in our work, during the period April – October, the mean temperature is about 17.1 ºC in VVR and 19.5 °C in DR, and according to the climate maturity grouping , the growing season can be defined as 'Intermediate' and 'Hot', respectively. [Cunha and Richter(2020)]
[Cunha and Richter(2016)], compare the average climates of two periods, 1950–1999 and 2000–2049 for the 27 most important wine regions and their results suggest that temperatures during the growing season could increase by an average of 1.3 °C. The greatest increase in temperature is projected to occur in Portugal. As they point in their work, some of those regions are at or close to their optimum growing season temperature for high quality and and WPV, and thus, this potential increment on the temperatures mean serious difficulties to the Grapevine

2.5 Climate based approaches for wine prediction

Many of the impact assessments of climate conditions on wine yield have been carried out based on statistical models supported by long-term time series of WPV and climate (e.g. Gouveia et al., 2011; Santos et al., 2011; Bock et al., 2013; Bonnefoy et al., 2013; Cunha and Richter, 2016; Teslić et al., 2019). However, there is still very little evidence how sensitive is the temporal variability of WPV to the scenarios that combine technological progress and actual or future climate variability. [Cunha and Richter(2020), Cunha and Richter(2016)]

In [Cunha and Richter(2016)] the impact of climate change on crop yield has been simulated using general or regional models as well as the crop-growth-monitoring systems. These were then combined with stochastic weather generators to analyse the impact of climate change on crop yield.

Other study based on autoregressive model [Cunha and Richter(2012)], or measures of radial grape growth rings, point towards identifiable induced-climate cyclically of WPV.

In [?] the cyclical proprieties of WPV in DR and VVR are modelled using a time-frequency approach based on Kalm filter regressions for a long term period close to 80 years in both regions. The short time Fourier transform was used to decompose the link between WPV and ST and SW. It is shown how many WPV cycles in each region, and what cycle in particular, are explained by the ST and SW.

A multi-variable mixed linear model was used to assess the relationship between various yield components and weather conditions in [Zhu et al.(2020)]

[Santos et al. (2011)] have developed a multivariate linear regression model of grapevine yield (1986-2008) in response to monthly mean temperatures and monthly precipitation totals from March to June for the DR. [Zhu et al.(2020)]
[Bock et al. (2013)] have analysed the correlation between long-term (1805–2010) grapevine yield records in Lower Franconia, Germany at the mean May-August temperature and sugar content at the mean April- August temperature. However, they did not consider the physiological basis of the grapevine yield formation, nor the effects of weather events during key developmental stages on yield. [Zhu et al.(2020)]

Linear mixed-effects model (lmer) from the R package of ‘lme4’ (Bates et al., 2014) was used to assess the relationship between different weather factors and yield component [Zhu et al.(2020)]. Using data from a long-term yield monitoring experiment with meteorology data, this study quantified the relationship between grapevine yield components (bunch number per vine, berry number per bunch, berry mass, bunch mass and yield per vine) and weather conditions during critical periods of grapevine development.

2.6 Distribution Rules

Many studies to assess the impact of climate change on crop yield, have assumed an historical probability distribution (e.g. [Vossen and Rijks (2001)]; [Santos et al. (2010)]; [Lorenzo et al. (2013)].
The use of such distributions assumes stationarity and a constant linear trend that in some way, describes the overall long-term trend in wine production. These approach's , does not reflect the implicit structural breaks in production [Quiroga and Iglesias(2009), Cunha and Richter(2016)]
To learn and discover distributions, [Jorge et al.(2006)] propose the use of distribution rules (DR). DR associate a frequent item-set with an empirical distribution of a numeric attribute of interest without information loss . In general, DR can be used in descriptive data mining tasks with the advantage of avoiding prediscretization of the numeric variable of interest.

DR can also potentially be used in a predictive setting, and are not fundamentally limited to numeric properties of interest.
Machine learning has focused particularly on learning conditional probabilities of one target variable y (either numerical or categorical) with respect to a set of input variables X. However, the output of a learning algorithm is typically reduced to associating the most adequate value of y to each combination of values of the variables in X.

[Jorge et al.(2006)] provide an algorithm that discovers distribution rules and how to filter interesting ones. Also, using the statistical distribution of Kolmogorov-Smirnov they show that DR can be easily visualized as frequency polygons and viewed by a domain expert or data analyst.
In this work we will use the [Jorge et al.(2006)] fundamental concepts and methodology's described about the processes of generating the rules, ways to filtering and presenting them within our wine production problem and also provide interest measures, and evaluation.

2.6.1 Conceptual framework

[Jorge et al.(2006)], approach the problems of discovering and presenting important conditional distributions of a target variable with respect to a set of input variables based on association rule discovery.

Association rules (AR) are highly legible chunks of knowledge that can be discovered from data. The process for generating association rules is efficient enough to deal with very large databases, although, our dataset is relatively small regarding the concept of large datasets in our days.
Although the result is very well defined for descriptive purposes, AR can also be useful in classification [Liu et al.(1998)], clustering [Han et al.(1997)], regression [Ozgur et al.(2004)], recommendation and subgroup discovery [Kavšek and Lavrač(2006a)].

Typically, algorithms for the discovery of AR deal with categorical attributes only. [Srikant and Agrawal(1996)] proposed a specific approach for the discretization of numerical attributes bearing in mind the descriptive aim of AR, recommendation and subgroup discovery [Kavšek and Lavrač(2006a)].
Distribution rules can be naturally applied subgroup discovery tasks, both for numeric and categorical properties of interest.
One immediate advantage of their use in these situations is that it is not required to previously discretize the attribute Y

Accordingly to [Jorge et al.(2006)], since the consequent of one distribution rule is an empirical distribution, it can be represented as a frequency polygon.
In their work, a visualization is suggested 3 where antecedent of each rule (e.g., the leftmost) is displayed as the main title.
Some selected measures of the distribution and the name of the property of interest (P.O.I) are shown within the plot. The x axis is the domain of the P.O.I. and the y axis the estimated probability density. The polygon is drawn by binning the domain of the P.O.I. into a given number of intervals.

The distribution for the set of cases that satisfy the condition is shown in black, and the default distribution for the whole population is shown in grey.
In the case of association rules, objective interest measures typically try to assess how much the observed frequency of the consequent of the rule, under conditions imposed by the antecedent, deviate from the frequency that would be expected assuming that antecedent and consequent were independent.

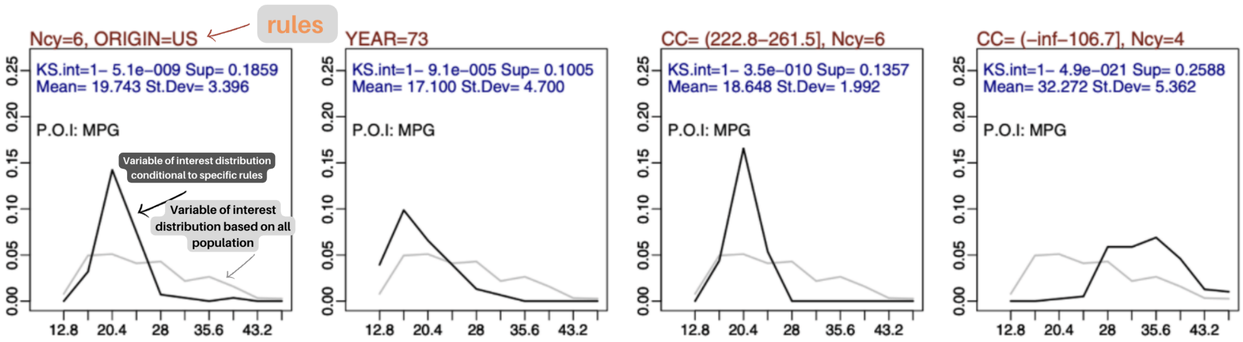

Figure 3: Rules Discovery example adapted from [Jorge et al.(2006)].
Rules discovery visualization approach

The difference between two empirical distributions can be assessed through a statistical goodness of fit test. [Jorge et al.(2006)] use the Kolmogorov-Smirnov test of the interest of a distribution rule, and filter a set of rules by selecting the ones with KS-interest above a pre-defined threshold.

2.6.2 Definition

Distribution rules are related to quantitative association rules but can be seen as a more fundamental concept, useful for learning distributions.
Distribution rules are mainly related to learning probability distributions [Kearns et al.(1994)], subgroup discovery and quantitative association rules (QAR).

Aumann and Lindell's work on QAR uses a z-test to identify rules significance [Webb et al.(2011)], pointed that, z-test is inappropriate for small samples.
The OPUS-IR authors propose the use of the standard t-test to decide the significant rules, arguing that the t-test tends to the z-test as the number of degrees of freedom increases.

For [Jorge et al.(2006)], both z-test and t-test assume normality which cannot be guaranteed. In this sense, they argue that, the Kolmogorov-Smirnov approach is an advantage since no further distribution assumptions is assumed. Also, with classical association rules a pre-discretize process to the numerical attribute of interest was required, but quantitative association rules, reduce the set of values in the consequent to a summary given by the mean or median.

Following and because our intention is to follow [Jorge et al.(2006)] method, we can define in a more formal way:

Definition 1 *A distribution rule (DR) is a rule of the form $A- > y = D_{y|A}$, where A is a set of items as in a classical association rule, y is a property of interest (the target attribute), and $D_{y|A}$ is an empirical distribution of y for the cases where A is observed. This attribute y can be numerical or categorical. $D_{y|A}$ is a set of pairs $y_j/freq(y_j)$ where y_j is one particular value of y occurring in the sample and freq(y_j) is the frequency of y_j for the cases where A is observed.*

In [Jorge et al.(2006)], y is a numeric variable. Nevertheless, the concept of distribution rules is extended for categorical attributes as well. The attributes on the antecedent are either categorical or are discretized as in [Jorge et al.(2006), Fayyad and Irani(1993a)]

2.6.3 Numeric Attributes of Interest

In [Jorge et al.(2006)] the algorithm CAREN-DR is used and it is shown that it works by finding frequent item-sets and, simultaneously, their associated p.o.i. distributions
As already seen, the algorithm uses a Kolmogorov-Smirnov test to select significant rules.
It is shown that the CAREN-DR algorithm when performed on 4 different datasets, with different values of minimal support for a minimal KS-interest of 0.95, it scales up well and is capable of generating a very large number of distribution rules.

In order to test the ability of the KS-test to identify the interesting rules, they used an artificial and a real dataset to evaluate the algorithm performance.
In the real case scenario, they have applied distribution rules to the analysis of the main causes of delays in trip time duration for buses in a urban centre. A dataset with 8000 cases describing trips of a specific bus line and 16 attributes plus the p.o.i that was the time of each Trip. Their were able to obtain 36 relevant rules. For example, the difference in the distribution of the time a bus takes to make its route if it is in March and August were a distinct distribution rule.
In this work, we will follow the same approach, by identifying all the possible attributes from the meteorological time series, that are related with the grapevines cycles, to extract some features, to be used to detect distribution rules, that could describe some of the variability of WPV.

2.7 Subgroup discovery

Subgroup discovery (SGD) in time series, refers to the process of identifying and extracting patterns or groups of similar time series within a dataset.
Variety of techniques that can be used for sub group discovery in multi - time series, include dynamic time warping, decision trees, association rule mining, and clustering algorithms.
SGD works by constructing a model that can explain the behavior of the time series data and then searching for subgroups of instances that have similar behavior within the model.

The goal of this data mining approach, is to find patterns or regularities to summarize the data. An important characteristic of this task is the combination of predictive and descriptive induction. [Herrera et al.(2011)]
As an example, predictive techniques maximise accuracy in order to correctly classify new objects, and descriptive techniques simply search for relations between unlabelled objects. The need for obtaining simple models with a high level of interest led to statistical techniques which search for unusual relations[Herrera et al.(2011)]
SGD is somewhere halfway between supervised and unsupervised learning. It can be considered that lies between the extraction of association rules and the obtaining of classification rules. [Usama et al.(1996)]

The visualisation of results is also very important for the usability of the knowledge extracted. This algorithms are often used to present the results to an expert, who will take decisions based on these data. In [Atzmüller and Puppe(2005)], different visualisation methods supporting exploratory and descriptive data mining were presented.

Another issue to be dealt with in more depth is the scalability. Many of the algorithms have a high computational cost when they are applied to large datasets. [Herrera et al.(2011)]. That could not be our case because as referenced our data is not to large. Nevertheless, it is a key point to take into consideration when evaluate possible candidates.

2.7.1 Conceptual framework

Subgroup discovery presented has a goal that is to generate in a single and interpretable way subgroups to describe relations between independent variables and a certain value of the target variable. [Usama et al.(1996)]
The algorithms for this task must generate subgroups for each value of the target variable [Herrera et al.(2011)]

Using domain knowledge in data mining methods can improve the quality of data mining results. In SGD, it can help to focus the search on the interesting subgroups related to the target variable by restricting the search space. Different approaches to include domain knowledge have been presented recently:

- Semantic Subgroup Discovery
- Domain Knowledge

There are several typoes of algorithms to be evaluated. Classification of the algorithms for subgroup discovery developed so far and identified in this review:

- **Extensions of classification algorithms:** EXPLORA [Usama et al.(1996)]; MIDOS [Wrobel(1997)] ; SubgroupMiner ; SD [Gamberger and Lavrac(2002)] ; CN2-SD [Lavrac et al.(2004)] ; RSD [Lavrac et al.(2002), Železnỳ and Lavrač(2006)]

- **Extensions of association algorithms:** APRIORI-SD [Kavšek and Lavrač(2006b), Kavšek et al.(2003)] ; SD4TS [Mueller et al.(2009)] ; SD-MAP [Atzmueller and Puppe(2006)] ; DpSubgroup [55] ; Merge-SD [54] ; IMR [Boley and Grosskreutz(2009)]
- **Evolutionary algorithm:** SDIGA [Del Jesus et al.(2007)] ; MESDIF [Berlanga et al.(2006), del Jesus et al.(2007)]; NMEEF-SD [Carmona et al.(2009), Carmona et al.(2010)]

One of the most important aspects in SGD is the choice of the quality measures employed to extract and evaluate the rules The most common quality measures used in subgroup discovery are described here, classified by their main objective such as complexity, generality, precision, and interest. [Herrera et al.(2011)]

2.7.2 Definition

The concept of subgroup discovery was initially introduced by Kloesgen [Usama et al.(1996)] and Wrobel [Wrobel(1997)], and more formally defined by Siebes [Siebes(1995)] but using the name Data Surveying for the discovery of interesting subgroups.
In subgroup discovery, we assume we are given a so-called population of individuals (objects, customer, ...) and a property of those individuals we are interested in. The task of subgroup discovery is then to discover the subgroups of the population that are statistically "most interesting", i.e. are as large as possible and have the most unusual statistical (distributional) characteristics with respect to the property of interest.[Herrera et al.(2011)]
As an example from [Herrera et al.(2011)]:

> Let *D* be a dataset with three variables *Age = Less than 25, 25 to 60, More than 60*, *Sex = M, F* and *Country = Spain, USA, France, German*, and a variable of interest, or target variable, *Money = Poor, Normal, Rich*.

Some possible rules containing subgroup descriptions could be:

R1: (Age = Less than 25 AND Country = German) $\rightarrow$ Money = Rich

R2: (Age = More than 60 AND Sex = F) $\rightarrow$ Money = Normal

Where rule R1 represents a subgroup of German people with less than 25 years old for which the probability of being rich is unusually high with respect to the rest of the population, and rule R2 represents that women with more than 60 years old are more likely to have a normal economy than the rest of the population.

Subgroup discovery in time series is a process of finding coherent, meaningful and interesting subgroups of time series data that exhibit specific patterns or characteristics (this can be useful in various applications such as customer segmentation, predictive maintenance, and fault detection).
Some of the algorithms used for subgroup discovery in time series include:

- **Rule-based methods:** These methods use predefined rules to identify subgroups in time series data. Examples include association rule mining and decision tree-based approaches.
- **Clustering methods:** These methods divide the time series data into different groups based on similarity measures. Examples include k-means, hierarchical clustering, and density-based clustering.
- **Feature-based methods**: These methods use feature extraction and dimensionality reduction techniques to identify subgroups in time series data. Examples include principal component analysis, independent component analysis, and singular value decomposition.

2.7.3 Concepts and Techniques

Subgroup discovery [Usama et al.(1996), Wrobel(1997)] is a broadly applicable data mining technique aimed at discovering interesting relationships. The patterns extracted are normally represented in the form of rules and called subgroups.
Because the specific characteristic of our data, an inspection upon the main characteristics of the algorithms is needed to foster a efficient selection to be tested. Based on the article from [Herrera et al.(2011)], an exhaustive research was made and there is sufficient information to initialise some kind of pre-selection of algorithms candidates. Here we present some of the most relevant concepts for each one.

The first algorithms developed for subgroup discover, *EXPLORA* and *MIDOS*, are extensions of classification algorithms and use decision trees.

MIDOS [Wrobel(1997)] applies the *EXPLORA* approach to multi-relational databases and is goal is to discover subgroups of the target variable that have an unusual statistical distribution with respect to the complete population.

SubgroupMiner is an extension of *EXPLORA* and *MIDOS* and it is an advanced SGD system that uses decision rules and interactive search in the space of the solutions.

SD [Gamberger and Lavrac(2002)] is a rule induction system based on a variation of the beam search algorithms.

CN2-SD is a subgroup discovery algorithm obtained by adapting a standard classification rule learning approach *CN2* to subgroup discovery

RSD (Relational subgroup discovery) has the objective of obtaining population subgroups which are as large as possible, with a statistical distribution as unusual as possible with respect to the property of interest, and different enough to cover most of the target population. It is an upgrade of the *CN2-SD* algorithm which enables relational subgroup discovery.

SubgroupMiner is the first algorithm which considers the use of numerical target variables, though it is necessary to perform a previous discretisation.

APRIORI-SD is developed by adapting to subgroup discovery the classification rule learning algorithm *APRIORI-C*,

SD4TS [Mueller et al.(2009)] is an algorithm based on *APRIORI-SD* but using the quality of the subgroup to prune the search space.

SD-Map [Atzmueller and Puppe(2006)] is an exhaustive subgroup discovery algorithm that uses the well-known *FP-growth* method for mining association rules with adaptations for the subgroup discovery task.

DpSubgroup [Grosskreutz et al.(2008)] is a subgroup discovery algorithm that uses a frequent pattern tree to obtain the subgroups.

Merge-SD [Grosskreutz and Rüping(2009)] is a subgroup discovery algorithm that prunes large parts of the search space by exploiting bounds between related numerical subgroup descriptions. In this way, the algorithm can manage datasets with numeric attributes.

IMR [Boley and Grosskreutz(2009)] is an alternative algorithmic approach for the discovery of non-redundant sub-groups based on a breadth-first strategy.

Some of these algorithms like *APRIORI-SD* or *SD4TS* are obtained from the adaptation to subgroup discovery of the association rule learner algorithm *APRIORI*, but others like *SD-MAP*, *DpSubgroup* or *Merge-SD* are adaptations of *FP-Growth*.

All of them use decision trees for representation. Only *Merge-SD* and *SD-MAP* can handle numeric or continuous variables, and for the rest a previous discretisation is necessary.

Evolutionary algorithms for extracting subgroups can be approached and solved as optimisation and search problems. They imitate the principles of natural evolution in order to form searching processes.

SDIGA[Del Jesus et al.(2007)] is an evolutionary fuzzy rule induction system. The algorithm evaluates the quality of the rules by means of a weighted average of the measures selected.

MESDIF [Berlanga et al.(2006), del Jesus et al.(2007)] is a multi-objective genetic algorithm for the extraction of fuzzy rules which describe subgroups. The algorithm extracts a variable number of different rules expressing information on a single value of the target variable.

NMEEF-SD [Carmona et al.(2009), Carmona et al.(2010)] is an evolutionary fuzzy system whose objective is to extract descriptive fuzzy and/or crisp rules for the subgroup discovery task, depending on the type of variables present in the problem

The evolutionary algorithms proposed so far for the subgroup discovery task are based on the hybridisation between fuzzy logic and evolutionary algorithms, known as evolutionary fuzzy systems.[Herrera et al.(2011)] One important note already pointed earlier , is the fact that, some of the variables collected in the datasets used to apply subgroup discovery techniques are continuous variables and most of the subgroup discovery algorithms are not able to handle continuous variables.
In this case, several models follow a strategy of a previous discretisation using different mechanisms [Fayyad and Irani(1993b), Liu et al.(2002)].

On the other hand , some approaches can manage continuous variables in the condition of the rule (explaining variables) without the need of a previous discretisation. This is performed in the evolutionary algorithms proposed in [Carmona et al.(2009), Del Jesus et al.(2007)] using fuzzy logic [Siebes(1995)].

An approach presented in [Moreland and Truemper(2009)], called *TargetCluster*, discretises a continuous target variable before applying a subgroup discovery algorithm based on a clustering approach.

In [Umek et al.(2009)], a methodology for grouping different variables as a target variable was presented.

This proposal is based on clustering to separately find clusters as values of the target variable.

3 Time series of climate and wine data

The data that will be used, is from several meteorological stations. There will be available data for the two regions: Douro (DR) and Vinhos Verdes Wine regions (VVR).
The information is about daily meteorological data (temperature, soil water moisture, rain...) and about the Regional annual wine production within those regions.4
The Periods available will be: 1933 to 2022 for the DR and from 1941 to 2022 for the VVR.
At the time of writing this report, data was still being collected and organized and further visualizations and insights about it, will be given in the final thesis document.

Tmin

Julian	Date	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
01-Jan	1	0.0	-2.0	6.0	2.0	2.0	5.0	11.5	4.0	5	1.0	0.0	10.0	3.5
02-Jan	2	0.0	-2.0	5.5	0.0	4.0	0.0	3.0	4.0	6	0.0	1.0	9.5	2.5
03-Jan	3	0.0	-1.5	5.5	1.0	4.0	1.0	1.0	0.0	6.5	3.0	5.5	6.5	1.0
04-Jan	4	0.0	-1.0	10.0	-1.0	2.0	2.5	0.5	2.0	6	-1.0	3.0	7.5	6.0
05-Jan	5	0.0	-2.5	11.0	2.0	3.5	2.0	5.0	0.0	10	0.0	5.0	9.0	3.0
06-Jan	6	0.0	-2.5	6.5	2.0	2.0	4.5	0.0	0.0	5.5	1.0	6.5	9.0	3.5
07-Jan	7	0.0	-1.0	4.5	4.0	2.5	2.5	2.0	0.0	2	3.0	5.0	10.5	4.0
08-Jan	8	0.0	-1.0	4.0	3.0	3.0	0.0	3.5	1.5	2.5	6.0	9.0	11.0	4.0
09-Jan	9	0.0	-1.5	6.0	2.5	3.0	-3.0	5.5	1.5	5	2.0	3.0	14.0	3.5
10-Jan	10	0.0	-5.0	10.0	1.5	0.0	-1.0	2.0	2.0	3.5	4.0	2.0	9.0	-1.5
11-Jan	11	0.0	0.0	6.5	2.5	1.5	0.0	0.5	-1.0	6	4.0	2.5	12.5	-3.5
12-Jan	12	0.0	-1.0	6.5	2.5	1.5	-3.0	3.0	5.0	1	5.5	2.0	9.5	1.5
13-Jan	13	0.0	-4.5	7.5	1.5	1.5	2.0	-1.0	5.5	3	5.0	2.5	7.5	1.0
14-Jan	14	0.0	-0.5	8.0	4.0	4.0	-0.5	0.0	-5.0	4	6.0	2.0	5.5	1.5

Figure 4: Example of the tyope of data being collected
Daoly Meteorological data

3.1 Experimental Design

In terms of the experimental design , for the Distribution Rules [Jorge et al.(2006)] method , we must use the business knowledge form the readings to extract several features. The idea is to use the expertise to build knowledge to work as features in our dataset.
For now, it seems obvious that features regarding the following topics should be considered for different time frames and regions:

- Spring Temperature for different years in different moments;
- Soil water in several moments, like month before crop;
- Summer Temperature Amplitude for different years until several years into the past;
- Rainfall, several moments last 2 years before crop;
- Temperature, rainfall, soil water, etc, during the principals stages of the Ripening and Berry Growth;
- Meteorological values from the variables during the Grapevine Flower Formation different stages

From the information collected, using the data within a frequency domain could derive some interesting properties to take into account as possible features. This means that , in the very first main step, the process of features extraction will be done in several domains has time domain, frequency domain..

Some of the most important information regarding the business expertise was obtained. Here we present some examples. A detailed list will be compiled to support this critical step of features extraction:

- SW is clearly more important to WPV than ST when it comes to predicting WPV [Cunha and Richter(2016)]
- The in-season ST is directly correlated with WPV and could explain about 65 % of its inter-annual variability. Results suggests that actual ST in Douro is generally below the optimum level of the main grape temperature-dependent physiological processes related with crop yield that occurs during spring such as flowering development, "anthesis" and fruit-set [Cunha and Richter(2016)]
- High ST plays an important role in triggering the different phenological stages, with great impact on SW stress avoidance and, consequently on WPV [Cunha and Richter(2016)]
- ST and SW showed great capacity to explain the cyclicality of WP in the studied regions.
- High ST is negatively correlated with late frosts spells. The positive effect of in-season ST on WPV agree with the previous studies on the Douro region.
- The time-varying spectrum presented in 6, which is based on the autoregressive model DR, shows the dynamic characteristics of the WP. Over the entire frequency band, there are three distinctive peaks at frequencies 0.1, 1.1 and 2.5.
- In the same studies, using lag values from the variables, they found that the 5^th lag is significant at the 5% significance level.

- WPV is characterised by a long run trend (frequency 0.1) and two shorter cycles of 5.7 years (frequency 1.1) and 2.5 years (frequency 2.5). This means that the time span from one peak in WP to another is close to 6 and 2.5 years, respectively.[Cunha and Richter(2020)]. Also ST with lag 3 have significant impact on the WP, in opposition to the current ST that an increase in the previous STs has a negative impact on WPV.

- As can be seen from 5 the VVR power spectrum is towards the end of the sample characterised by three dominant frequencies: the long run trend, the frequency of 1.3 and 2.6, which corresponds to the 4.8 and 2.4 year cycles. [Cunha and Richter(2020)]

- the coherence between SW and WPV for the VVR. For the entire sample there are three distinctive frequencies visible: the long-run trend, the frequency of 1.2 (or 5.2 years), and the frequency of 2.2 (or 1.5 years). Both, the long-run trend as well as the 5.2 years cycles.

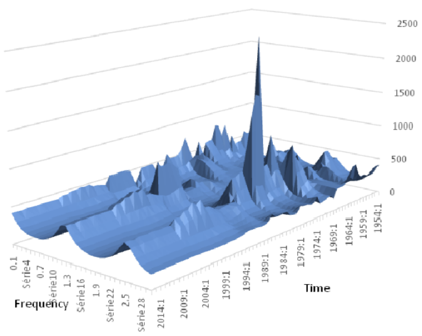

Figure 5: Time varying power spectrum for the wine production for the period 1937-2016 in Vinhos Verdes Region - Figure from [Cunha and Richter(2020)]

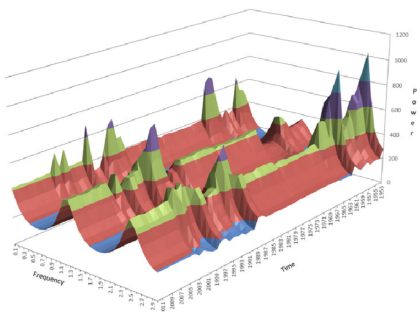

Figure 6: Time varying power spectrum for the wine production for the period 1933-2013 in Douro Region - Figure from [Cunha and Richter(2020)]

As for the Subgroup discovery part, some research was already done but it will be studied in more detail, how to select an appropriate algorithm and how to build the frame work and experimental design for that technique.

4 Work Plan and Roadmap

The expectations related with thus work is to be able to run all the experiments regarding the two technical topics, Sub Group Discovery (SGD) and Distribution Rules (DR) about the Wine Production and the meteorological impacts. Also, because there's no relevant information founded in the literature reviewed until this stage about the application of SGD regarding time series as features to explain a single point prediction variable, the expectations are directed oriented to the fact that research should be done and there is a risk of some of the work should be followed in future work. Nevertheless the work developed during this thesis will be of higher importance for that future work.
The DR methodology that will be followed, could be a straight forward application, and the main results will be directly related with the findings upon the data used and not so directed to research or innovation regarding the available literature.
Related with the work plan, a high amount of the literature was reviewed, but is expected that further readings would be needed especially regarding the SGD topic.
Regarding the business understanding, there is already sufficient information to create the necessary features, to use with the DR methodology. The current status about the work plan defined, the figure 7 gives a good perspective of the actual road map and next's steps and major milestones.

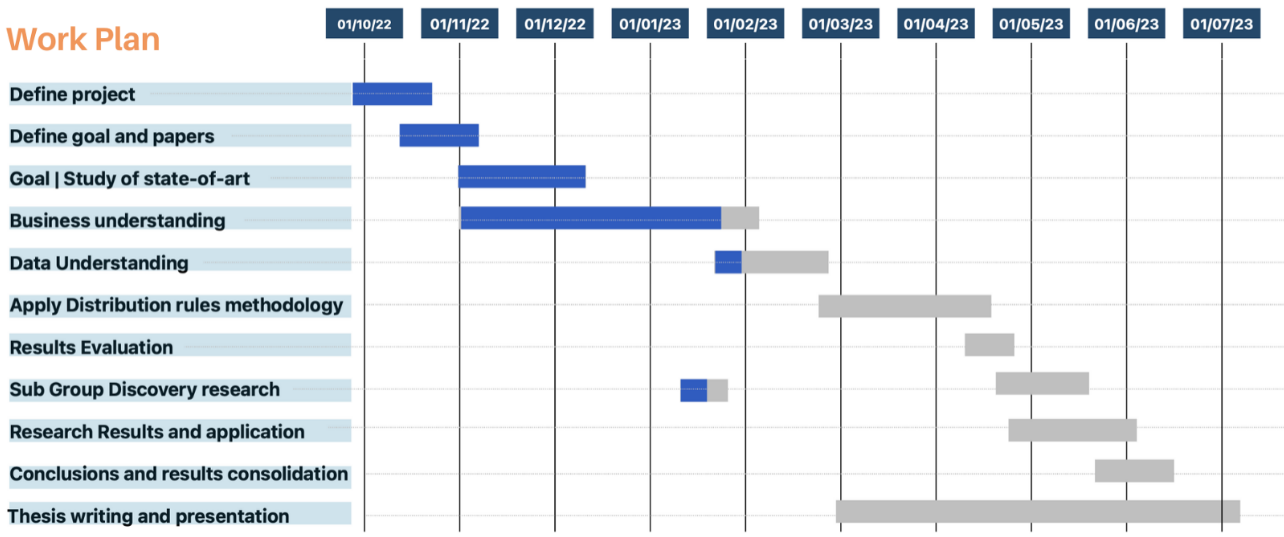

Figure 7: Road Map with the main milestones about the work plan

Road Map and main milestones

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